

JOEL HÄGVALL

Software Engineer | Open Source & Workflow Automation

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PROFILE

Software Engineer focused on TypeScript, developer tooling, workflow automation and reliable integrations. Currently building enterprise software at Scania and preparing Maskera, a local-first Swedish PII redaction toolkit, for an August 2026 open-source launch. Experienced in debugging APIs and third-party dependencies, authentication flows, automated testing, CI/CD and clear asynchronous documentation.

EXPERIENCE

Creator & Maintainer

Maskera | Public pre-release | Jun 2026 - Present | Open-source launch planned Aug 2026

Swedish-first, client-side PII redaction for browser, Node.js, edge and AI workflows.

Web: [maskera.dev](#) | npm: [npmjs.com/package/maskera](#) | Model: [huggingface.co/joelhagvall/maskera-sv-ner](#)

- Built the TypeScript packages maskera and @maskera/core; pre-release versions are available on npm ahead of the open-source launch.
- Trained, evaluated, quantized and published joelhagvall/maskera-sv-ner on Hugging Face; its q4 ONNX artifact is about 43 MB and detects Swedish names, locations, organizations and street addresses via Transformers.js.
- Designed a local-first architecture that combines checksum-validated rules for structured identifiers with on-device model inference, keeping raw text on the user's device.
- Built benchmark gates, release automation and synchronized pre-release documentation across npm, Hugging Face and a technical whitepaper; turned recurring edge cases into tests and detection improvements.

Software Developer

Scania | Sep 2024 - Present

- Build and maintain internal .NET/C# microservices and TypeScript/React tooling from investigation through implementation and support.
- Debug issues across application logic, integrations and third-party dependencies, following edge cases through to reliable fixes.
- Create workflows, automations and integrations in BMC Remedy across internal business processes.
- Work with AWS and cloud-related development tasks across internal systems.

Sole Software Engineer

ResiliaAI | Part-time, pre-launch venture | 2025 - Present

AI-powered resilience and crisis preparedness platform for housing associations and organizations.

- Sole engineer in a three-person team, building the platform alongside full-time work at Scania.
- Built and deployed the pre-launch platform end-to-end across development and hosted environments using React, TypeScript, Node.js, PostgreSQL/Supabase, REST APIs and cloud hosting.
- Developed AI workflows including RAG over 24,000+ vectorized Q&A entries, embeddings, vector search, agents and multi-provider LLM orchestration.
- Built authentication, admin tooling, audit trails and observability alongside Jest/Playwright test workflows, CI/CD and sharded end-to-end testing.

R&D Summer Intern

Ericsson | Summer 2023 & Summer 2024

- Developed Python tooling for radio performance verification teams.
- Debugged and improved Java software through bug fixes, SonarQube and SonarLint findings, and maintainability work.
- Used Jenkins in development and testing workflows; returned for a second summer after receiving a return offer.

COMMUNITY & LEADERSHIP

Vice Head of Marketing

Systemvetardagen | Sep 2023 - Mar 2024

- Led communication and coordination for a technology career fair with 3,000+ attendees, managing partner relationships, planning and team delivery.

SELECTED PROJECTS

Data Wipe Mailer

Built and launched a privacy-focused GDPR deletion tool with Next.js, TypeScript, Tailwind CSS and shadcn/ui. The launch generated 200,000+ LinkedIn views and 2,000+ likes.

Repo: github.com/joelhagvall/data-wipe-mailer | Live: data-wipe-mailer.vercel.app

J.A.R.V.I.S

Built a private local AI assistant for macOS with SwiftUI, Ollama and MCP servers for personal notes, tools and local-first AI workflows.

Repo: github.com/joelhagvall/jarvis-chat

Tor Onion Site Scraper

Built a Python crawler and analysis platform for a university thesis, retrieving Tor HTML pages, extracting marketplace data and supporting quantitative and qualitative analysis.

Repo: github.com/joelhagvall/tor-onion-site-scraper | Thesis: su.diva-portal.org

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, React, Vite, Next.js, Node.js, Express.js, .NET/C#, Java, Python

APIs & Automation: REST APIs, SDK integrations, API keys, authentication, authorization, OpenAPI checks, BMC Remedy workflows

Open Source & AI: npm, pnpm, Git, GitHub, Transformers.js, ONNX, Hugging Face, NER, local inference, Vercel AI SDK, RAG, MCP

Testing & Delivery: Jest, Vitest, Playwright, unit/integration/E2E testing, CI/CD, Jenkins, release automation, benchmark gates

Cloud & Data: AWS, Docker, PostgreSQL, Supabase, Neon, Railway, Vercel, logging and observability

EDUCATION

B.Sc. in Computer and Systems Sciences

Stockholm University | 2021 - 2024 | 180 ECTS